

Bowen Song

Engineering / MLOps / ML Systems

Research engineer across Red Hat OpenShift, eBay pipelines, and deployable ML systems.

ML systems break under scale, traffic shifts, and schema change, so I build platforms and pipelines that stay reliable through those failures.

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EXPERIENCE

Graduate Researcher, Personalized AI for Health

2020-Present

University of Southern California | Los Angeles, CA

Wearable sensing is noisy and deployment is constrained, so I built multimodal pipelines and personalized models that stay interpretable on device and useful across users.

- Research pipelines were fragmented across sensing modalities, so I built end-to-end data, training, and evaluation workflows that made comparison under noise and subject shift reliable enough to guide model decisions.
- One-size-fits-all models broke across users, so I framed personalization around latency, privacy, and federated adaptation to keep the system deployable instead of trapped in a lab setting.

PhD Research Intern, Core Ranking

2024

eBay | San Jose, CA

Ranking gains had to survive real traffic and strong baselines, so I built production-constrained model and evaluation workflows that made improvements worth trusting.

- Multi-objective ranking needed more selective capacity, so I designed Transformer MoE models that fused text and structured signals and created a scalable path to specialization.
- Traffic heterogeneity kept breaking naive expert routing, so I built distributed training and evaluation loops that preserved specialization gains under capacity limits and shifting demand.

PhD Research Intern, Infrastructure

2023

eBay | San Jose, CA

Real-time analytics depended on trustworthy ingestion, so I owned a Kafka-to-ClickHouse pipeline that kept data correct through exactly-once behavior and safer operations.

- Multi-datacenter ingestion risked duplicate or dropped events, so I integrated Kafka Connector into ClickHouse with exactly-once semantics and made streaming analytics safer to trust.
- Upgrades and schema changes could break downstream consumers, so I hardened rollout and evolution paths to keep the pipeline operable across environments instead of brittle during change.

Software Engineer, OpenShift

2019-2020

Red Hat (IBM) | Boston, MA

Production clusters need upgrades and debugging to be boring, so I shipped Kubernetes platform features and CI tooling that made operations safer and easier to trust.

- Operator installs and upgrades were fragile at cluster scale, so I owned OLM changes for dependency resolution, upgrade safety, and version orchestration that made cluster rollouts safer.
- Distributed build failures obscured real regressions, so I built statistical flake detection that made CI output faster to trust and faster to debug.

PROJECTS

KUBERNETES PLATFORM

Operator Lifecycle Manager

Cluster upgrades are easy to break, so I contributed to OLM to manage dependency-aware installs and safer lifecycle orchestration on OpenShift clusters.

CI RELIABILITY

Flake Analyzer

Flaky CI hides real regressions, so I built statistical detection and classification tooling that made distributed test failures faster to trust and debug.

LARGE-SCALE RECONCILIATION

RCDS_GO

Large remote states are expensive to resync, so I implemented content-aware reconciliation that sends differences instead of whole payloads and makes huge-file sync more practical.

SKILLS

Systems: Kubernetes, Operators, Helm, containers, and upgrade safety for cluster-scale software delivery; Kafka, ClickHouse, streaming data, exactly-once ingestion, and schema evolution for trustworthy pipelines; Cloud federation, QoS reasoning, benchmarking, and reproducible infrastructure for scalable shared systems

ML Platform: PyTorch, TensorFlow, distributed training, experiment pipelines, and GPU clusters for production-minded ML; Krylov, Ray, Kubeflow, and evaluation loops for large-scale production ML iteration

Languages: Python, Go, C/C++, Java, Scala, MATLAB, R, LaTeX

EDUCATION

PhD in Computer Science

University of Southern California | Los Angeles, CA | 2020-2026 (expected)

MS in Electrical and Computer Engineering

Boston University | Boston, MA | 2019

PUBLICATIONS

Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements

IEEE CLOUD 2022 | Bowen Song, Marco Paolieri, Leana Golubchik

Hybrid cloud sharing is hard to size and trust, and this paper provides a QoS and incentive model for deciding when federation is worth deploying.

Scalable String Reconciliation by Recursive Content-Dependent Shingling

Allerton 2019 | Bowen Song, Ari Trachtenberg

Synchronizing large remote states is bandwidth-heavy, and this paper introduces a reconciliation method that performs well on real revision histories and large files.